

COMPREHENSIVE PROJECT REPORT

Development of a Gazebo-ROS2 Simulation Framework for Cooperative Path Planning for an Autonomous Marine Vehicle

Workspace: /home/sireesha/marine_ws

Status: Phase 1-6 complete; Phase 7 in progress; Phases 8-9 planned

Prepared from the project execution history and retained reported results

October 2026

Project Phase Roadmap

Executive Summary

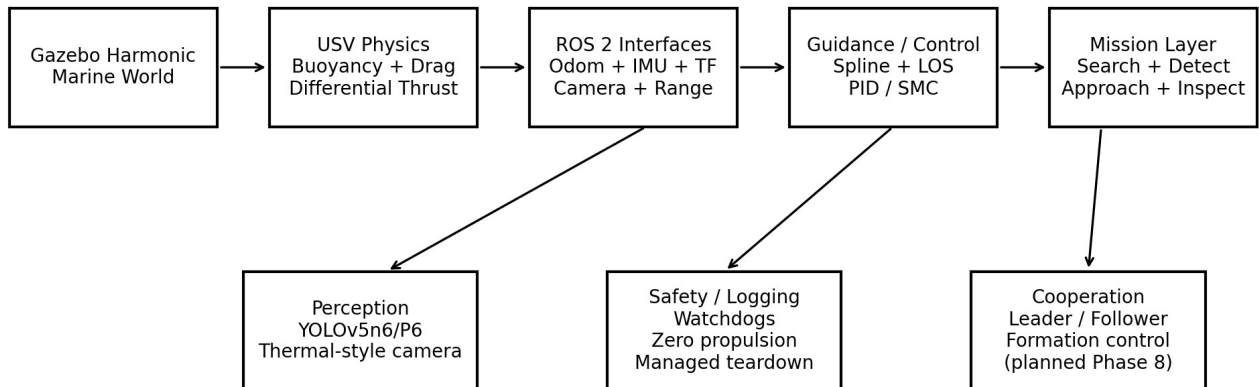
This report consolidates the full development path, completed tasks, quantitative results, safety work, remaining modules, and final execution plan for the ROS 2 / Gazebo Harmonic autonomous marine vehicle research framework. The project has progressed from a basic simulation world to a physically simulated catamaran USV with spline/LOS guidance, PID and Sliding Mode Control (SMC), AI vessel detection, autonomous search/approach/inspection, safety supervision, and disturbance validation. Cooperative two-vehicle control and final repository integration remain to complete the planned framework.

Current project state

Phases 1-6 are complete and frozen. Phase 7 has a validated disturbance plugin, 299 relevant tests, a successful strong-profile smoke test, and the straight-path PID moderate-disturbance trial completed; the straight-path SMC trial was interrupted by the Codex usage limit while still under the unchanged mission deadline. Phase 8 and Phase 9 have not started.

Area	Current status	Key result / next action
Simulation & vehicle physics	Complete	Gazebo Harmonic catamaran with buoyancy, drag and differential thrust
Guidance	Complete	Natural cubic-spline paths + LOS guidance
PID controller	Complete	Frozen $K_p=0.5$, $K_i=0$, $K_d=0.6$
Perception & mission	Complete baseline	YOLOv5n6/P6 integrated with search, range-confirmed arrival and inspection
SMC controller	Complete	Frozen $\lambda=0.15$, $k=0.05$, $\phi=0.08$, $\omega=0.5 \text{ s}^{-1}$
PID vs SMC comparison	Complete	18 post-correction attempts, 9 matched pairs; descriptive conclusions frozen
Disturbance robustness	In progress	Resume interrupted straight SMC, then curved/S-path pairs + limited sensor/actuator checks
Cooperative control	Planned	Two independent vehicles, leader/follower formation and communication robustness
Final integration	Planned	Three final scenarios, reproducibility docs, launch wrapper, final freeze

Framework Architecture (current + planned final integration)



1. Project Objectives and Scope

The project objective is to build a modular, reproducible marine autonomy simulation framework in ROS 2 and Gazebo Harmonic for development and evaluation of path-following, nonlinear control, AI-assisted vessel search, autonomous approach/inspection, and cooperative control. The project intentionally keeps simulation, guidance, control, perception, mission logic, safety supervision, and evidence generation as separable modules.

1.1 Core research goals

- Provide a physically simulated under-actuated surface vehicle with differential thrust and realistic-enough buoyancy/hydrodynamic damping for controller evaluation.
- Generate smooth straight, curved and S-shaped reference paths using cubic-spline interpolation and follow them with Line-of-Sight (LOS) guidance.
- Implement and compare a conventional PID heading controller and a mathematically derived SMC under matched conditions.
- Integrate a vessel detector into a simulated thermal-style camera pipeline and connect perception to search, target acquisition, approach and inspection logic.
- Use an independent range sensor for final arrival verification rather than target ground-truth coordinates.
- Demonstrate robust safe-stop behavior, stale-data handling, zero-propulsion verification and managed process teardown.
- Complete disturbed-water validation, then extend to two-vehicle cooperative control and freeze a runnable research framework.

1.2 Non-goals intentionally excluded

- No MPC, reinforcement learning, new controller families, or additional PID/SMC tuning after Phase 6 freeze.
- No retraining or replacement of the current YOLO architecture during finalization.
- No CFD-grade hydrodynamics, fluid-structure interaction, SLAM, hardware-in-the-loop, or large multi-vehicle swarm.
- No large statistical campaign beyond the retained matched experiments unless a specific defect invalidates existing evidence.

2. Software and Runtime Environment

Component	Configuration
Host	Windows 11 + WSL2 Ubuntu 22.04.5
ROS	ROS 2 Humble
Simulator	Gazebo Harmonic 8.15.0
Python	Python 3.10
Bridge	ros_gz_bridge / ros_gz_sim
Workspace	/home/sireesha/marine_ws
Legacy workspace	~/maritime_ws must remain untouched
GPU note	Existing NVIDIA configuration must remain unchanged; WSLg/software rendering may be used where needed

3. Phase-by-Phase Development Summary

Phase	Purpose	Status	Primary output
1	Simulation foundation	Complete	Packages, calm_ocean.sdf, clock bridge
2	Physical USV	Complete	Catamaran dynamics, thrusters, odom/IMU/TF
3	Guidance	Complete	Cubic splines and LOS
4	PID path following	Complete	Calm-water benchmark
5A-H	AI perception + mission reliability	Complete baseline	Search/detect/approach/range/inspect + safety corrections

Phase	Purpose	Status	Primary output
6	SMC and PID comparison	Complete & frozen	Selected SMC + 9 matched post-correction pairs
7	Disturbance/sensor/actuator robustness	In progress	Plugin + smoke + principal disturbed campaign
8	Two-vehicle cooperation	Planned	Leader/follower formation
9	Final integration and freeze	Planned	Runnable framework + final reports

4. Phase 1 - Simulation Foundation

Phase 1 established the ROS 2 / Gazebo workspace structure and a stable calm-water simulation baseline.

- Created the core packages `marine_description`, `marine_gazebo` and `marine_bringup`.
- Created `calm_ocean.sdf` and the baseline world/vehicle launch structure.
- Established the simulation clock bridge and basic ROS-Gazebo connectivity.
- Validated WSLg/Gazebo startup and preserved the existing host/NVIDIA environment.

5. Phase 2 - Physical USV and Propulsion

- Introduced the catamaran USV as a physically simulated surface craft.
- Added graded buoyancy and body-frame hydrodynamic drag.
- Implemented differential-thrust propulsion and the existing velocity-to-thrust interface.
- Exposed `/asv/cmd_vel`, odometry, IMU, and TF feedback for closed-loop control.
- Preserved the same physical propulsion pathway for all later PID and SMC experiments.

6. Phase 3 - Path Generation and LOS Guidance

- Implemented natural cubic-spline interpolation for smooth reference trajectories.
- Created straight, curved and S-shaped test paths.
- Implemented LOS guidance to compute desired heading and cross-track error.
- Kept the guidance layer independent of the inner-loop controller so PID and SMC could be compared fairly.

7. Phase 4 - PID Control Baseline

The existing controller is a heading PD/PID-form controller with $K_i=0$. Its gains were frozen as the historical baseline and were not subsequently retuned.

Parameter	Value
Kp	0.5
Ki	0.0
Kd	0.6

Path	Historical CTE RMSE (m)	Historical heading RMSE (rad)	Mission time (s)
Straight	0.569	0.083	93.392
Curved	0.574	0.078	105.912
S-shaped	0.583	0.084	132.279

Interpretation

These Phase 4 results are historical reference values. Phase 6 correctly used fresh matched PID trials against SMC rather than treating the older benchmark as the primary comparison dataset.

8. Phase 5 - AI Perception, Search, Mission Integration and Reliability

8.1 Detector and dataset characterization

- The supplied best.pt and last.pt checkpoints were identified as a custom YOLOv5n6/P6 10-class model, not YOLOv5s.
- The model contains approximately 3.11 million parameters, with detection heads at strides 8/16/32/64 and 640-pixel inference size.
- An IR vessel archive contained roughly 9,000 grayscale images split 7,200/900/900. The archive data.yaml class order conflicted with the embedded checkpoint class names, so the project intentionally avoided unsupported subclass-accuracy claims.
- Smoke tests produced detections on 29/30 IR test images. Initial RGB/grayscale/CLAHE Gazebo images did not trigger useful detections.
- An assigned-temperature thermal-style rendering approximation produced vessel-overlapping detections in 20/20 test images at approximately 0.308-0.316 confidence with the original best.pt threshold of 0.25. This is a simulation approximation, not calibrated infrared imaging.

8.2 Search and integrated mission

- Implemented a three-lane coverage search and retained two successful diagnostic search runs around 205.5 s with CTE RMSE near 0.557 m.
- Created marine_perception and marine_mission packages and connected camera, detector, mission state, guidance and control.
- Implemented detection confirmation, a visual-heading/reference arbiter, independently range-validated arrival, and inspection logging.
- Kept target ground truth out of autonomous navigation; it is reserved for offline evaluation only.

8.3 Phase 5H root-cause and reliability work

Checkpoint	Key finding / action
5H-1	Audit showed 4,924/4,924 recorded nonempty detections reached downstream during primary 5G; valid-range availability had fallen to ~79.15%, and accepted-pair accounting was doubled.
5H-2	Controlled comparison isolated an extra full-resolution depth observer as a major range-delivery impairment: observer OFF ~93.37% valid range versus ON ~81.34%. Applied corrections_v1: observer default OFF and accepted-pair counter fix.
5H-3	Lean instrumentation showed no downstream guidance/PID/thrust-mixer defect; two integrated missions inspected successfully with complete detection propagation.
5H-4	Thirteen bounded fault injections verified safe recovery/zero propulsion; native clock/teardown issues remained separate.
5H-4B	Managed teardown investigation found controllers-first ordering removed observed native failures in 9/9 managed trials. Applied teardown_fix_v1.
5H-5	Final 10-trial reliability campaign: 8 inspections, 9 clean shutdowns, 7 combined passes, two arrival timeouts and one post-inspection PID SIGSEGV; mean valid-range availability 92.19%; zero propulsion verified in all ten.

Phase 5 conclusion

The perception/mission stack is demonstrably functional but not perfect. Failures were retained rather than tuned away. Mission outcome, propulsion safety, and native process outcome are treated as separate dimensions.

9. Phase 6 - Sliding Mode Control and Matched PID Comparison

9.1 SMC mathematical design

The design audit found that commanded yaw rate cannot be treated as actual yaw rate because the existing mixer includes drag feedforward. The SMC therefore computes a yaw-moment demand using nominal inertia/damping and maps it through the original bounded yaw-rate interface.

$$\begin{aligned} \mathbf{s} &= \mathbf{v}_f - \mathbf{r} + \lambda \cdot \text{wrap}(\psi_f - \psi) \\ \tau^* &= 10\mathbf{r} + 20|\mathbf{r}| \mathbf{r} + J_0[\mathbf{a}_f + \lambda(\mathbf{v}_f - \mathbf{r}) + k \text{sat}(\mathbf{s}/\phi)] \\ \mathbf{q} &= \text{clip}[-0.4, 0.4](h^{-1}(\tau^*)), \quad h(\mathbf{q}) = 10\mathbf{q} + 20|\mathbf{q}|\mathbf{q} \end{aligned}$$

SMC parameter	Frozen value
λ	0.15
k	0.05
ϕ	0.08
Reference bandwidth ω	0.5 s^{-1}
Nominal inertia J_0	45.993 kg m^2
Yaw-rate command limit	$\pm 0.4 \text{ rad/s}$

9.2 SMC tuning and validation

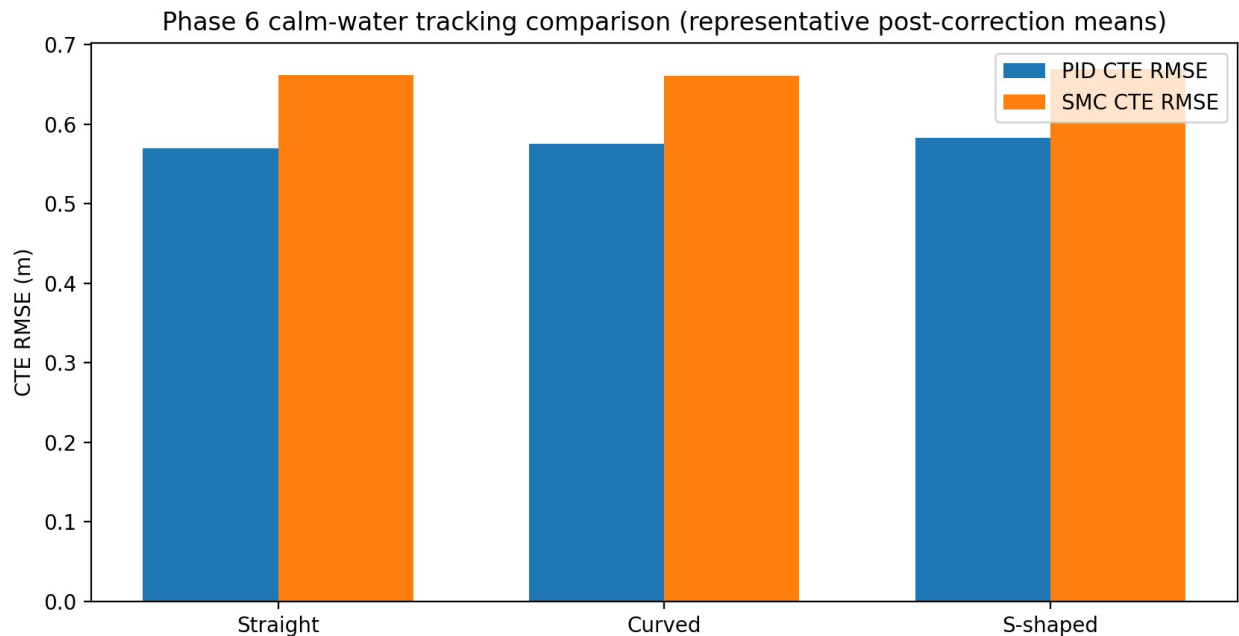
$\lambda / k / \phi$	Straight CTE RMSE (m)	Heading RMSE (rad)	Observation
0.10 / 0.03 / 0.05	0.708	0.0922	Conservative baseline
0.15 / 0.03 / 0.05	0.662	0.0857	Improved tracking
0.15 / 0.05 / 0.05	0.643	0.0807	Best straight-path tracking
0.15 / 0.05 / 0.08	0.660	0.0850	~60% lower tail command variation with small tracking penalty

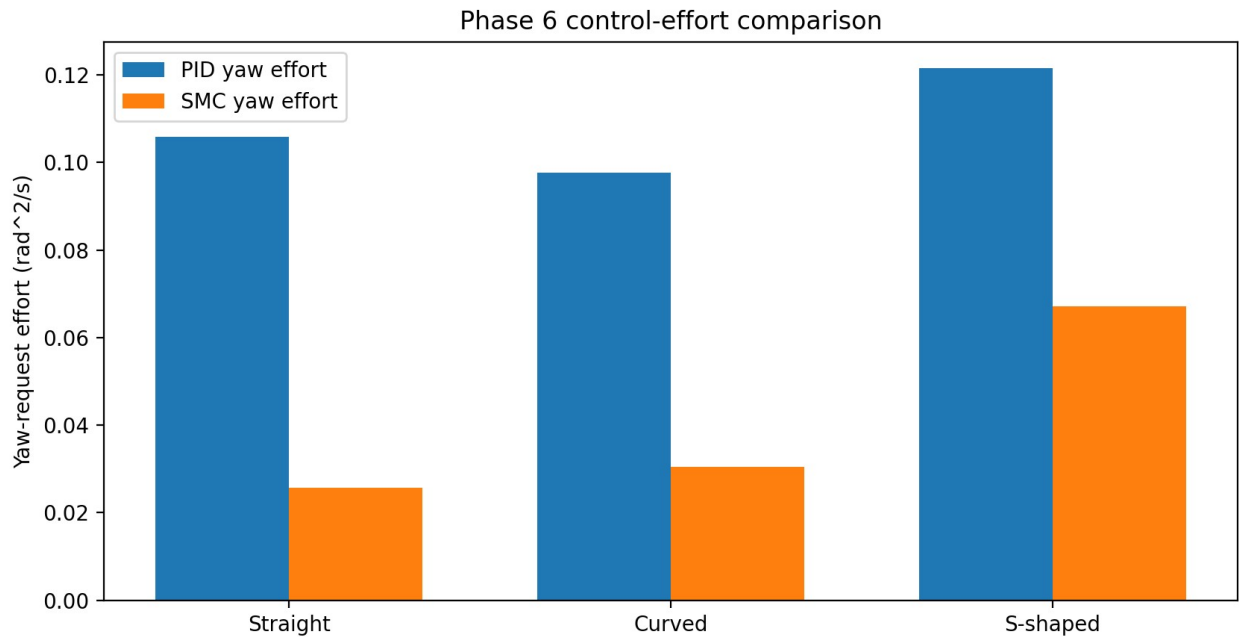
The S-shaped path initially exceeded the predefined 0.10 rad tail-heading RMSE limit. A single controlled change increased the reference-model bandwidth from 0.4 to 0.5 s^{-1} while keeping $\lambda/k/\phi$ and all bounds unchanged. Tail heading RMSE improved from 0.12040 rad to 0.09014 rad, with an independent repeat at 0.09010 rad. Straight and curved finalization trials also passed, so the selected configuration was frozen separately without changing the original default SMC YAML.

9.3 Final matched PID-versus-SMC comparison

Phase 6 was finalized using the available post-correction Batch 2-4 cohort only. Batch 5 was planned but was not executed. The final primary dataset contains 18 post-correction attempts, corresponding to nine matched PID/SMC pairs (three pairs per path).

Metric	Final observation
Mission completion	18/18
Verified zero propulsion	18/18
Process success	17/18
PID tracking	Lower CTE and tail heading RMSE across all nine matched pairs
SMC effort	Lower yaw-request effort across all nine matched pairs
SMC command behavior	Greater command variation
Saturation	0% in retained post-correction trials
Statistical claim	Descriptive only; no significance claim





9.4 Timestamp-ordering correction

- Batch 1 exposed three stale_odometry mission failures where odometry stamps were 1-4 ms ahead of controller time.
- Both PID and SMC adapters had treated any negative age as stale.
- A shared causal selector was implemented: future samples remain pending until simulation time reaches their timestamp, while the latest still-fresh causally eligible sample is retained.
- The original 0.5 s watchdog deadline was not loosened.
- Offline validation: 283 tests passed (218 existing + 65 new).
- Live validation: 402 injected 1-4 ms samples were handled causally; genuine expiry still triggered safe zero propulsion. A natural 1 ms pending event was later handled correctly in Batch 2.

10. Phase 7 - Disturbance and Robustness Validation (In Progress)

10.1 Completed work

- Designed an optional deterministic Gazebo disturbance plugin that applies bounded world-frame force and yaw moment per physics step.
- Disturbance is seed- and simulation-time-driven so PID and SMC can receive identical profiles independent of message delivery timing.
- Calm behavior remains unchanged when the plugin is disabled.
- Documented disturbance levels and principal-trial plan.
- Initial build was contaminated by Windows Anaconda libraries appearing in WSL PATH; this was resolved using a Linux-only separate build/install directory without changing system packages or Phase 6 artifacts.
- Plugin build succeeded and 299 relevant tests passed, including 16 new profile tests.

- Strong-profile smoke validation passed: all 30,785 physics-step wrench samples matched the deterministic formula; causal feedback, exclusive command authority, bounded outputs, verified zero thrust and clean teardown all passed.
- First principal moderate-disturbance straight PID trial completed with CTE RMSE about 1.329 m, zero saturation, verified zero propulsion and clean teardown. It failed the original calm-quality thresholds, as expected under the disturbance, and the failure was retained.

10.2 Interrupted straight SMC trial

The matching straight SMC moderate-disturbance trial reached end-of-path progress but had not yet satisfied the unchanged arrival-radius condition when the Codex session hit its weekly usage limit. The experiment was allowed to continue under the original deadline; on resumption, the first task is to inspect retained evidence and determine whether a valid final outcome already exists before rerunning anything.

10.3 Remaining Phase 7 tasks

1. Inspect and classify the interrupted straight SMC result; rerun only if the session interruption prevented a valid experimental outcome.
2. Run the remaining predetermined curved PID/SMC and S-shaped PID/SMC moderate-disturbance trials serially.
3. Calculate degradation versus frozen Phase 6 calm baselines, including CTE, heading, effort, saturation, endpoint acquisition and process outcomes.
4. Run short sensor robustness diagnostics: bounded latency, short dropout, out-of-order samples, small future timestamps and clock pause/resume.
5. Run limited actuator diagnostics: symmetric thrust reduction, asymmetric port/starboard reduction and temporary saturation.
6. Finalize docs/phase7_disturbance_report.md and docs/evidence/phase7/; update README; stop before Phase 8.

11. Phase 8 - Two-Vehicle Cooperative Control (Planned)

Phase 8 will add the cooperative-control contribution without reopening the PID/SMC comparison. The simplest defensible architecture is a leader/follower pair using independent vehicle instances and a lightweight coordinator.

Task	Required behavior
Two-vehicle launch	Independent Gazebo models, namespaces, odometry, TF, cmd_vel, thrusters and safe stop; no topic/TF collisions
Leader	Follow existing cubic-spline/LOS path and publish cooperative state such as path progress, pose, heading and velocity
Follower coordinator	INITIAL_TRACKING -> WAIT_FOR_THRESHOLD -> FORMATION_ACQUIRE -> FORMATION_HOLD -> COMMUNICATION_DEGRADED -> COMPLETE/SAFE_STOP
Formation	Use one fixed documented lateral or longitudinal offset;

Task	Required behavior
	engage only after predefined tracking thresholds are satisfied
Communication robustness	Bounded latency and short dropout; never use indefinitely stale cooperative state; recover or enter safe state
Experiments	Exactly three principal missions: straight/low-curvature, curved, S-shaped
Metrics	Leader/follower CTE, heading errors, inter-vehicle distance error, formation RMSE/max error, time to formation, completion and zero propulsion for both vehicles

12. Phase 9 - Final Integration, Reproducibility and Framework Freeze (Planned)

- Preserve all completed algorithms; no new controller, planner or neural-network architecture is introduced.
- Run limited perception robustness checks: reduced thermal-style contrast, moderate image noise, partial target visibility and temporary detection dropout.
- Validate search/approach robustness: delayed detection, target loss/reacquisition, intermittent range and short range-sensor dropout while keeping mission deadlines unchanged.
- Run exactly three final live scenarios: (A) frozen single-USV path-following, (B) full search -> YOLO detection -> confirmation -> approach -> independent range-confirmed arrival -> zero propulsion -> inspection, and (C) two-USV cooperative mission.
- Create a requirements traceability matrix that links every research claim to implementation files, experiment evidence, quantitative metric and known limitation.
- Perform one final clean-shell build and relevant regression suite.
- Create reproducible launch entypoints and, if useful, a thin wrapper such as scripts/run_marine_demo.sh with basic/pid/smc/disturbance/perception/mission/cooperative modes.
- Create docs/reproducibility.md, docs/final_project_report.md, docs/final_results_summary.md and docs/final_manifest.json.
- Finalize README as the project entry page and mark the repository COMPLETE - RUNNABLE RESEARCH FRAMEWORK.

13. Safety, Reliability and Evidence-Integrity Principles

- Exactly one controller is authoritative for /asv/cmd_vel during a trial.
- Future-dated odometry may be buffered but is never used before it becomes causally eligible.
- Genuinely stale feedback still triggers the unchanged 0.5 s watchdog and safe zero propulsion.
- Mission completion, propulsion-safety success and process-cleanup success are reported separately.
- Native teardown crashes occurring after verified zero propulsion are preserved as process failures rather than rewritten as tracking failures.

- Failures are retained and classified; they are not automatically rerun simply to obtain passing results.
- Historical evidence is protected with preservation audits and hash manifests.
- Phase 6 frozen parameters and Phase 5H corrections must remain unchanged during Phases 7-9.

14. Package / Module Architecture

Package / area	Role
marine_description	Vehicle model and description assets
marine_gazebo	Gazebo world/simulation integration and physical environment
marine_bringup	Launch orchestration, phase launch files and managed startup/shutdown
marine_control	PID, SMC, odometry feedback selection and control interfaces
marine_guidance	Cubic-spline reference paths and LOS guidance
marine_perception	Camera bridge and YOLO vessel-detection pipeline
marine_mission	Search, confirmation, guidance arbitration, range-confirmed arrival and inspection state logic
scripts/	Experiment supervisors, diagnostics, metrics, plotting, preservation and reproduction tools
docs/evidence/	Raw and processed experimental evidence organized by phase

15. Comprehensive Task Checklist

Phase	Task	Status
P1	Workspace/package skeleton	Complete
P1	Calm ocean world	Complete
P1	Clock bridge	Complete
P1	Basic launch validation	Complete
P2	Catamaran physics	Complete
P2	Buoyancy/drag	Complete

Phase	Task	Status
P2	Differential thrusters	Complete
P2	Odom/IMU/TF	Complete
P3	Natural cubic splines	Complete
P3	Straight/curved/S paths	Complete
P3	LOS guidance	Complete
P3	CTE/desired-heading outputs	Complete
P4	PID node	Complete
P4	Frozen gains	Complete
P4	Three-path benchmark	Complete
P4	Metrics/logging	Complete
P5	YOLO model characterization	Complete
P5	Thermal-style simulation	Complete
P5	Detector ROS integration	Complete
P5	Coverage search	Complete
P5	Mission state machine	Complete
P5	Range-confirmed arrival	Complete
P5	Inspection	Complete
P5	5H reliability root-cause	Complete
P5	Depth-observer fix	Complete
P5	Accepted-pair accounting fix	Complete
P5	Managed teardown correction	Complete
P5	Final reliability campaign	Complete
P6	SMC model audit	Complete
P6	Mathematical controller	Complete
P6	Unit tests	Complete
P6	ROS integration	Complete

Phase	Task	Status
P6	PID/SMC selector	Complete
P6	SMC tuning	Complete
P6	Reference bandwidth correction	Complete
P6	Frozen selected_v1	Complete
P6	Matched PID/SMC campaign	Complete
P6	Causal odometry fix	Complete
P6	Offline/live validation	Complete
P6	Final analysis/freeze	Complete
P7	Disturbance design/plugin	Partial
P7	Profile tests	Partial
P7	Smoke validation	Partial
P7	Straight PID disturbed trial	Partial
P7	Straight SMC final outcome	Pending
P7	Curved PID/SMC	Pending
P7	S-shaped PID/SMC	Pending
P7	Sensor robustness	Pending
P7	Actuator robustness	Pending
P7	Phase 7 report	Pending
P8	Dual vehicle namespacing	Pending
P8	Leader state sharing	Pending
P8	Follower coordinator	Pending
P8	Formation acquisition/hold	Pending
P8	Communication latency/dropout	Pending
P8	Three cooperative missions	Pending
P8	Phase 8 report	Pending
P9	Perception degradation checks	Pending

Phase	Task	Status
P9	Target loss/reacquisition	Pending
P9	Range dropout checks	Pending
P9	Three final scenarios	Pending
P9	Traceability matrix	Pending
P9	Clean build/regression	Pending
P9	Runnable entrypoints	Pending
P9	Reproducibility guide	Pending
P9	Final reports/manifest	Pending
P9	README freeze	Pending

16. Exact Completion Roadmap

Shortest defensible route

Resume Phase 7 from retained evidence -> finish disturbed PID/SMC trials -> run limited sensor/actuator diagnostics -> finalize Phase 7 -> implement two-vehicle cooperative control in Phase 8 -> perform Phase 9 final integration/reproducibility -> freeze the repository. No Phase 10 is required.

- Resume Phase 7 by inspecting the interrupted straight SMC evidence before rerunning anything.
- Complete the remaining disturbed-water matched trials and limited robustness diagnostics.
- Implement and validate the two-vehicle leader/follower cooperative architecture with three principal missions.
- Run final perception/search/range robustness checks and exactly three final integrated scenarios.
- Perform one final clean build/regression, verify launch commands, generate reproducibility/final reports, and freeze the framework.

17. Definition of Done - Runnable Research Framework

Capability	Final expected state
Basic simulation	One documented launch starts the Gazebo marine world and USV
Navigation	Straight/curved/S paths run with cubic spline + LOS

Capability	Final expected state
Controller choice	PID or frozen SMC selectable without competing publishers
Disturbance	Optional reproducible disturbance disabled by default
Perception	Existing YOLOv5n6/P6 pipeline runs on thermal-style camera imagery
Autonomous mission	Search -> confirm -> approach -> independent range stop -> inspection
Cooperation	Two independent vehicles execute leader/follower formation mission
Safety	Causal odometry, stale watchdog, verified zero propulsion, managed teardown
Evidence	CSV/logs/plots/reports retained under docs/evidence by phase
Reproducibility	Clean-shell build and documented launch commands validated
Project status	COMPLETE - RUNNABLE RESEARCH FRAMEWORK

18. Known Limitations and Future Work

- The thermal-style image pipeline is a simulation approximation and is not a calibrated thermal/infrared sensor model.
- The YOLO checkpoint class mapping has historical ambiguity, so mission logic correctly treats it as generic vessel detection rather than making unsupported subclass-accuracy claims.
- Phase 6 controller results are descriptive with three post-correction matched pairs per path, not a large statistical campaign.
- PID reference joins have a retained scalar/stamped-heading coherence limitation; missing joins were not silently filled.
- Native Gazebo/bridge teardown failures may still occur independently of propulsion safety and are reported separately.
- The SMC stability argument is conditional/local under model uncertainty and available control authority; no global sampled-data proof is claimed.
- Disturbance robustness is currently in progress and cooperative-control evidence has not yet been generated.
- Real-world sea trials, hardware-in-the-loop, CFD-grade hydrodynamics, multi-sensor fusion, SLAM, MPC/RL and larger swarms remain appropriate future work rather than current completion requirements.

Appendix A - Representative Build and Run Workflow

```
cd /home/sireesha/marine_ws
source /opt/ros/humble/setup.bash
# build using the project's validated build process
source install/setup.bash
```

Phase 9 will finalize the exact clean-build command, demo launch family, and a reproducibility guide. Until then, experiment supervisors and launch files should be used from the existing retained phase documentation rather than inventing new commands.

Appendix B - Final Status Snapshot

Dimension	Status
Core ROS/Gazebo platform	Complete
Physical USV	Complete
Spline + LOS	Complete
PID	Complete/frozen
AI perception/search/inspection	Complete baseline
SMC	Complete/frozen
Matched calm PID vs SMC	Complete
Causal odometry safety	Complete and live-validated
Disturbance robustness	In progress
Sensor/actuator stress	Pending within Phase 7
Two-vehicle cooperation	Pending Phase 8
Final integrated runnable framework	Pending Phase 9

End of report.